

Hate Speech Detection

➤ Associated paper:

**Dascalu, S., Hristea, F., Towards a Benchmarking System for Comparing Automatic Hate Speech Detection with an Intelligent Baseline Proposal. Mathematics 2022, 10, 945.
<https://doi.org/10.3390/math10060945>**

- **Hate Speech** is a frequent problem occurring among Internet users. Thus:
 - ✓ automatic Hate Speech detection is a very important topic for major social media platforms.
- We explore different transformer and LSTM-based models in order to evaluate the performance of multi-task and transfer learning models used for Hate Speech detection.
- **Our conclusion**: transformer-based models have the best performance on all studied Datasets.

➤ **Definition of Hate Speech**

- **Various definitions of Hate Speech exist. We use here the less constraining U.K.'s definition of Hate Speech.**
 - **The U.K.'s definition removes the burden of demonstrating the incitement to violence. The United Kingdom forbids communication of hatred that is directed towards someone on the basis of that person's racial, nationality, or belief characteristics.**
- **Hate Speech can be very hard to distinguish from Offensive Language.**

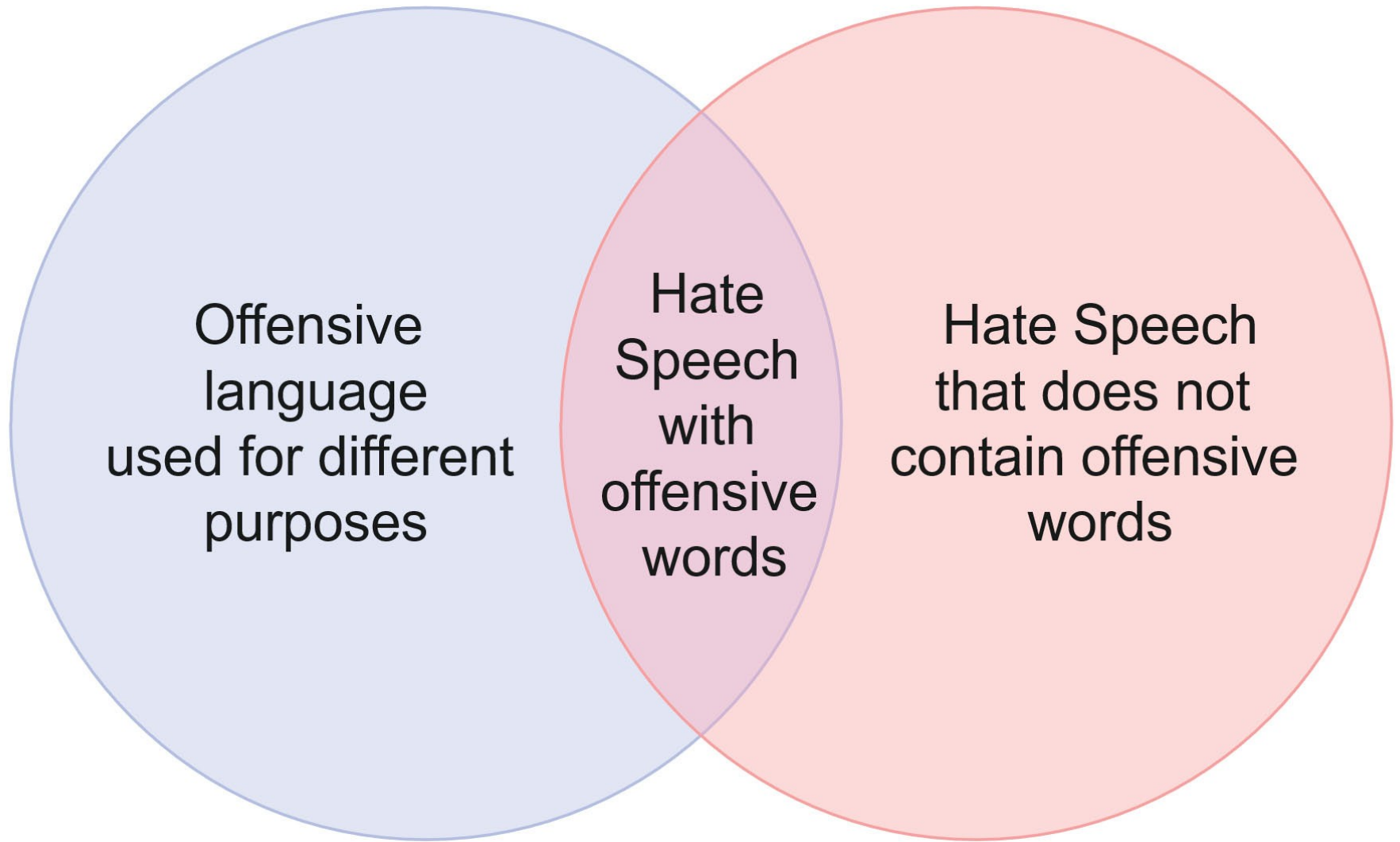


Figure 1. Usage types for Hate Speech key words

- **Recent research concerning Hate Speech shows that keywords are not enough in order to classify its usage.**
- **The semantic analysis of the text is required in order to isolate other usages that do not have the intention or meaning required for the phrase to be classified as Hate Speech.**

- **The topic of Hate Speech detection is a very active part of the NLP field:**
- ✓ **Every modern advance in textual classification has been subsequently followed by an implementation in the field of Hate Speech detection.**

- **Early methods relied on classic machine learning methods such as SVMs, logistic regression, Naive Bayes, decision trees, and random forests.**
- **Even earlier research was focused on different ways of extracting textual features and text representations such as n-grams, lexicon, and character features.**
- **These laid a foundation for later research, which evolved with other related tasks, such as sentiment evaluation, culminating with the usage of transformer-based neural networks.**

- **Modern approaches are highlighted by the use of embeddings and neural networks capable of processing sequences of tokens.**
- **Another modern technique is using one Dataset to improve performance on the primary Dataset (transfer learning).**
- **An interesting model presented in the literature is parameter-based sharing. This method distinguishes itself by using embeddings that are learned from secondary tasks and are combined with the primary task embeddings based on a learnable parameter. This allows the model to adjust how the secondary task contributes to the primary one, based on the relatedness of the tasks.**
- **Multiple recent papers make use of the transformer architecture in order to achieve higher accuracies.**

Our study: wants to determine if **sentiment analysis Datasets**, which by far outweigh Hate Speech ones in terms of data availability, can help in training for our Hate Speech detection problem.

➤ **For this purpose: we use multi-task learning.**

- **Multi-task learning represents the method that leverages multiple learning tasks, trained at the same time, with the objective of exploiting the resemblances of the Datasets when training the parameters of the models.**
- **When to use multi-task learning: this method is used when the task is comprised of a very large Dataset for the secondary task and the primary (objective) task Dataset does not have sufficient data to produce an accurate result when trained by itself.**

The datasets that are used in the experiments



Figure 2. Dataset label distribution

Types of Embeddings Used in the Experiments

- **TF-IDF**
- **Skipgram**
- **CBOW**
- **GloVe**
- **ELMo**
- **RoBERTa**

Evaluation

- Hate Speech predictors focus on the macro F1 metric (the harmonic mean of the recall and precision metrics).
- The sampling procedure most used to evaluate classification models on a segmented Dataset is K-fold cross-validation.
- Here: we chose 10-fold because it is the most popular for Hate Speech classification.

Preprocessing

The proposed pipeline for these kinds of models was, after preprocessing, composed of the following word embedding extraction methods:

- **N-gram char level features;**
- **N-gram word features;**
- **POS of the word;**
- **TF (raw, log, and inverse frequencies)-IDF (no importance weight, inverse document frequency, redundancy);**
- **No lemmatization running words were directly taken as input - no lemmatization, but unlemmatizable words were deleted;**
- **GloVe;**
- **ELMo;**
- **Transformers.**

Optimizers

An optimizer is an algorithm that helps the neural network converge, by modifying its parameters, such as the weights and learning rate, or by employing a different learning rate for every network weight.

Tested optimizers:

- SGD
- Adam
- RMSProp
- AdaGrad

Models

In order to test the methods, we propose the pipeline in Figure 3:

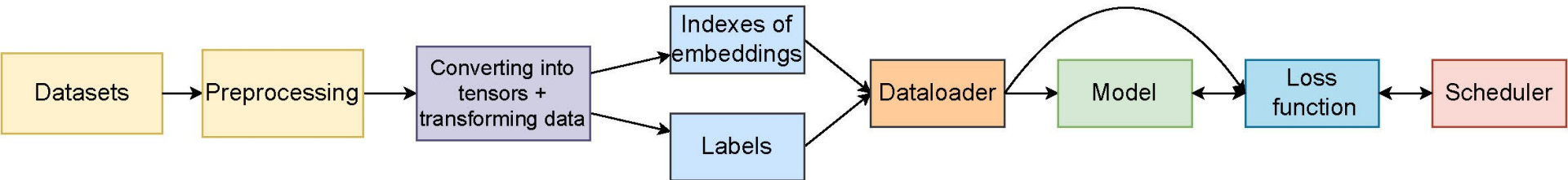


Figure 3. High-level architecture of the testing pipeline

Tested models

- **CNN, GRU, and RNN Model**
- **LSTM Based Networks**
- **Multi-Task Learning Models**
- **Learnable Parameter Multi-Task Learning Models (a multi-task learning model that has a learnable parameter; can lead to performance gain or not)**
- **Transformer-Based Models (using the BERT model in 3 architectural ways; BERT token embeddings for single-task learning; RoBERTa token embeddings for multi-task learning)**

Results

- **The previously mentioned models were tested on each selected Dataset.**
- **Test results can be seen in Tables 1–5.**
- **These tests used only the textual features of the samples.**
- **The results were rounded to the nearest hundredth (so the approximation error was 0.005).**
- **The numerical results of all the performed experiments (see Tables 1–5) point to the fact that using transformer models yielded better accuracies overall than using token embeddings.**

Table 1. BERT token embeddings

Dataset	Accuracy	Precision Macro	Precision Weighted	Recall Macro	Recall Weighted	F1 Macro	F1 Weighted
Xplain	0.67	0.66	0.67	0.66	0.67	0.66	0.67
Waseem	0.84	0.81	0.84	0.78	0.84	0.79	0.84
Kurrek	0.86	0.86	0.87	0.86	0.86	0.86	0.86
Gao–Huang	0.75	0.69	0.74	0.68	0.75	0.68	0.74
Hate	0.91	0.78	0.9	0.74	0.91	0.74	0.9
Davidson	0.91	0.78	0.90	0.72	0.91	0.74	0.90
Qian-Reddit	0.91	0.89	0.91	0.88	0.91	0.88	0.91

Table 2. RoBERTa-based transformer model

Dataset	Accuracy	Precision Macro	Precision Weighted	Recall Macro	Recall Weighted	F1 Macro	F1 Weighted
Xplain	0.7	0.69	0.7	0.69	0.7	0.69	0.7
Waseem	0.86	0.81	0.86	0.82	0.86	0.81	0.86
Kurrek	0.89	0.89	0.89	0.89	0.89	0.89	0.89
Gao–Huang	0.77	0.73	0.77	0.71	0.77	0.71	0.77
Hate	0.91	0.79	0.91	0.75	0.91	0.76	0.91
Davidson	0.91	0.77	0.91	0.75	0.91	0.75	0.91
Qian-Reddit	0.91	0.88	0.92	0.89	0.91	0.88	0.92

Table 3. MTL (fixed parameter) RoBERTa token embeddings

Dataset	Accuracy	Precision Macro	Precision Weighted	Recall Macro	Recall Weighted	F1 Macro	F1 Weighted
Xplain	0.64	0.63	0.64	0.63	0.64	0.63	0.64
Waseem	0.84	0.81	0.84	0.78	0.84	0.78	0.84
Kurrek	0.84	0.84	0.84	0.84	0.84	0.84	0.84
Gao–Huang	0.75	0.70	0.75	0.70	0.75	0.69	0.75
Hate	0.90	0.75	0.89	0.72	0.9	0.73	0.89
Davidson	0.9	0.75	0.89	0.7	0.9	0.71	0.89
Qian-Reddit	0.91	0.88	0.91	0.88	0.91	0.88	0.91

Table 4. MTL (learned parameter) RoBERTa token embeddings

Dataset	Accuracy	Precision Macro	Precision Weighted	Recall Macro	Recall Weighted	F1 Macro	F1 Weighted
Xplain	0.65	0.64	0.65	0.64	0.65	0.63	0.64
Waseem	0.84	0.81	0.85	0.79	0.84	0.79	0.84
Kurrek	0.84	0.84	0.84	0.84	0.84	0.84	0.84
Gao–Huang	0.75	0.71	0.75	0.67	0.75	0.67	0.74
Hate	0.9	0.77	0.9	0.72	0.9	0.73	0.9
Davidson	0.9	0.76	0.89	0.70	0.9	0.71	0.89
Qian-Reddit	0.91	0.88	0.91	0.88	0.91	0.88	0.91

Table 5. MTL (hard) RoBERTa token embeddings

Dataset	Accuracy	Precision Macro	Precision Weighted	Recall Macro	Recall Weighted	F1 Macro	F1 Weighted
Xplain	0.65	0.64	0.65	0.64	0.65	0.64	0.65
Waseem	0.84	0.81	0.84	0.79	0.84	0.79	0.84
Kurrek	0.84	0.85	0.85	0.84	0.85	0.84	0.84
Gao–Huang	0.75	0.71	0.74	0.65	0.75	0.65	0.72
Hate	0.9	0.77	0.9	0.73	0.9	0.73	0.9
Davidson	0.9	0.76	0.89	0.69	0.9	0.72	0.89
Qian-Reddit	0.91	0.88	0.91	0.88	0.91	0.87	0.91

Conclusions

➤ Our objectives were:

- establishing a Dataset list with different features, available for testing and comparisons;
- testing multi-task learning, and to determine how it compares with single-task learning with modern transformer-based models in the case of the Hate Speech detection problem.

➤ Results:

- using transformer models yielded better accuracies overall than trying token embeddings;
- when speed is a concern, then using token embeddings is the preferred approach, as training them used less resources and yielded a comparable accuracy;
- using auxiliary Datasets helped the models achieve higher accuracies only if the Datasets had the same range of classification (sentiment analysis or Offensive Language detection were complementary for Hate Speech), but also the same textual features (using tweet Datasets for longer text Datasets can be detrimental to its convergence).